

Day 2 Write-Up: Preprocessing & First Supervised Models

UCI Adult (Census Income) | Binary income prediction | NetixSol Internship

1. Preprocessing Plan & Choices

Feature split. Numeric features (6): age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week. Categorical features (8): workclass, education, marital-status, occupation, relationship, race, sex, native-country. Missing values were handled in Day 1 by adding an 'Unknown' category, so no NaN remained; the pipelines still include imputers to stay robust on new data.

Numeric pipeline: median imputation + StandardScaler. Median was chosen over mean because capital-gain (max 99,999) and capital-loss are heavily right-skewed; the median is robust to such outliers whereas the mean is pulled by them. StandardScaler is required by logistic regression, whose gradient-based updates are sensitive to feature scale.

Categorical pipeline: most-frequent imputation + OneHotEncoder. OneHotEncoder was chosen over LabelEncoder because label encoding imposes a meaningless ordinal ranking (is 'Never-married' < 'Married'?), which misleads linear models. One-hot creates independent binary columns per category. `handle_unknown='ignore'` protects against categories in the test set that were never seen in training.

Alternatives considered/skipped: mean imputation (outlier-sensitive), dropping rows with missing values (loses data), label/ordinal encoding (false ordering), dropping categorical features entirely (loses information).

Why treat preprocessing inside the pipeline? Fitting the imputer/scaler on the train set only and reusing those fitted transformers on the test set prevents **data leakage**. If statistics from the test set leaked into training-level imputation/scaling, the held-out evaluation would be optimistically biased.

2. Model Comparison on Hold-Out Test

Model	Acc	Prec	Rec	F1	ROC AUC	PR AUC
Day1 Rule (edu-num>=13)	0.753	0.484	0.497	0.491	0.665	0.361
Logistic Regression	0.854	0.743	0.598	0.663	0.906	0.767
Decision Tree (default)	0.819	0.617	0.639	0.628	0.757	0.481

Confusion matrices: LR `[[TN=6947, FP=484],[FN=940, TP=1398]]`, Tree `[[TN=6502, FP=929],[FN=843, TP=1495]]`. For LR, false negatives (940) exceed false positives (484) — it misses high earners more than it over-predicts, which softly constrains recall. For the tree, false positives (929) exceed false negatives (843) — it over-predicts the positive class, which hurts precision, our primary metric. Since the business goal is avoiding wasted outreach, a model with fewer over-predictions (LR) aligns better with the objective.

3. Interpretability Findings

Logistic Regression (interpretable coefficients): strongest positive drivers are capital-gain (+2.26), Married-civ-spouse (+1.54), executive/managerial occupation (+0.82), and education-num (+0.72). Strongest negative drivers are private-household-service (-1.46), Never-married (-1.29), and being female (-1.03) — the last is a known bias in this census data that should be flagged, not hidden.

Decision Tree: depth 72, 5,668 leaves, train accuracy 99.99% vs test 81.86% — a gap of 0.18 confirms severe overfitting. Top-level splits (is married? then capital-gain, then education) are domain-sensible, so the tree's logic is reasonable; it just memorizes the training set when allowed to grow unconstrained.

4. Model Selection for Day 3

Continuing with: Logistic Regression. It delivers the highest precision (0.743 vs 0.617 for the tree and 0.484 for the Day 1 rule baseline) plus the best ROC AUC (0.906) and PR AUC (0.767). For a campaign whose cost driver is wasted outreach, precision is the deciding metric and LR is the strongest on it while remaining fully interpretable.

The decision tree remains a secondary candidate: it is non-linear and reaches higher recall (0.639 vs 0.598), catching some high earners LR misses. But it overfits badly and its lower precision makes it less aligned with the business objective unless constrained. Day 3 will (1) test preprocessing changes — drop `fnlwgt`, collapse native-country to a US/Other flag, drop the redundant education column, group marital-status, bin/log-skewed capital-gain/loss, and cap hours-per-week outliers; (2) revisit the tree/Random Forest with depth constraints to compare fairly; and (3) keep LR as the primary development target.